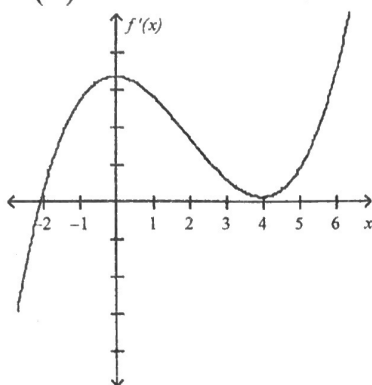


Multiple Choice: Write the CAPITAL LETTER corresponding to the correct answer on the line provided
A 1. Below is the graph of $f'(x)$. For what value(s) of x does $f(x)$ have a local minimum?



- A. $x = -2$ B. $x = 0$ C. $x = 4$ D. None

B 2. Let $f(x)$ be a continuous function. If $f'(x) > 0$ for $x > c$ and $f'(x) < 0$ for $x < c$ then what type of critical point is $(c, f(c))$?

- A. Local maximum C. Neither a local max nor local min
 B. Local minimum D. Unknown

C 3. For what values of x is $f(x) = x + \frac{1}{x}$ decreasing?

- A. $-1 < x < 1$
 B. $x < -1$

- C. $-1 < x < 0, 0 < x < 1$
 D. all real numbers

E 4. If $f(x) = -2x^3 - 6x^2 + 8$, then which of the following statements is true?

- I. $f'(x) > 0$ for $x \in (-2, 0)$
 II. $f'(-2) = f'(0) = 0$
 III. $f(x)$ has one point of inflection

D 5. A critical number is:
 A. I only B. I and II only C. III only D. II and III only E. I, II & III

- A. A number c in the domain of a function f such that $f'(c) > 0$.
 B. A number c in the domain of a function f such that $f'(c) < 0$.
 C. A number c in the domain of a function f such that $f'(c) = 0$.
 D. A number c in the domain of a function f such that $f'(c) = 0$ or does not exist.
 E. None of the above

6. True (T) or False (F).

- (a) If local max point is $(2, -9)$, then $f'(2) = 0$ FALSE. Could be a cusp or corner
 (b) A function has an absolute maximum at a if $f(a) \geq f(x)$ for all x in the domain. True.
 (c) A local maximum point can also be an absolute maximum point but not vice-versa. omit

7. Find the absolute extreme values of the function $g(x) = x^{\frac{2}{3}}(5+x)$, if $-5 \leq x \leq 1$.

$$g'(x) = \frac{2}{3} x^{-\frac{1}{3}}(5+x) + x^{\frac{2}{3}}$$

$$= \frac{2}{3} x^{-\frac{1}{3}} \left[5+x + \frac{3}{2}x \right]$$

$$= \frac{2}{3} x^{-\frac{1}{3}} \left[5 + \frac{5x}{2} \right]$$

$$g'(x) = 0 \text{ or DNE}$$

$$\Rightarrow x = -2 \text{ or } 0.$$

$$g(-5) = 0 \rightarrow \text{absolute min.}$$

$$g(1) = 6 \rightarrow \text{absolute max.}$$

$$g(0) = 0 \rightarrow \text{absolute min.}$$

$$g(-2) = 4.76$$

8. For Given the function $y = f(x)$ as shown, state

a) the interval(s) where

i. $f'(x) < 0$ & $f(x) > 0$ $(-3, -1.6) \cup (2, 4)$

ii. $f'(x) > 0$ & $f(x) < 0$ $(-1, 0)$

b) the local extrema values (specify the "type")

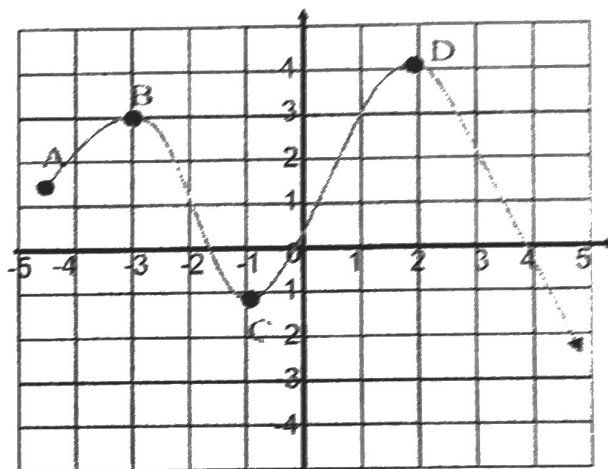
Local min: -1

Local max: 3, 4

c) the points where $f'(x) = 0 \rightarrow$ B, C, D

d) absolute maximum: 4

e) absolute minimum: $-\infty$



9. Find the intervals on which the function $f(x) = (x^2 - 9)^{\frac{2}{3}}$ is increasing and the intervals on which it is decreasing. Determine the x-coordinates of all local extrema.

$$f'(x) = \frac{2}{3} (x^2 - 9)^{-\frac{1}{3}} (2x) = \frac{4}{3} x (x^2 - 9)^{-\frac{1}{3}}$$

$\therefore$ critical #'s: 0, ± 3

	$(-\infty, -3)$	$(-3, 0)$	$(0, 3)$	$(3, \infty)$
x	-	-	+	+
$(x^2 - 9)^{\frac{1}{3}}$	+	-	-	+
$f'(x)$	-	+	-	+
$f(x)$	↓	↑	↓	↑

$\therefore$ Increasing: $(-3, 0) \cup (3, \infty)$

Decreasing: $(-\infty, -3) \cup (0, 3)$

Local min: @ $x = -3, 3$

Local max: @ $x = 0$